Explainable Accounting-Fraud Detection with Machine Learning

Extending Bao, Ke, Li, Yu & Zhang (Journal of Accounting Research, 2020)

Minor project · scale-free features, tuned ensembles, decision metrics, calibrated scores and SHAP explanations

What this talk covers

1 Paper highlights

What Bao et al. (2020) did and found

2 Literature review

Data, protocol, metrics, benchmarks, limitations

3 Research gaps

Five concrete things the paper leaves open

4 Our implementation

Features, models, evaluation rigour, deployment

5 Results

All numbers, with confidence intervals

6 Future work

Realistic major-project extensions

1 · The research paper

Bao, Ke, Li, Yu & Zhang (2020), "Detecting Accounting Fraud in Publicly Traded U.S. Firms Using a Machine Learning Approach", Journal of Accounting Research 58(1). About 330 citations.

Paper highlights and key summary

28

raw accounting items beat 14 expert ratios

0.72

AUC of RUSBoost on 2003-2008 test years (as reported, approx.)

2 yr

gap between training and test years (fraud is found late)

NDCG@k

new metric: quality of the TOP of the ranking

- Problem: predict which U.S. public firms will be charged with accounting fraud (SEC AAERs) one year ahead.
- Method: RUSBoost, boosting plus random under-sampling, built for rare events (0.66% of firm-years are fraud).
- Inputs: 28 raw Compustat items chosen from accounting theory, instead of hand-made ratios.
- Benchmarks: Dechow et al. (2011) F-score logistic model and Cecchini et al. (2010) SVM with a financial kernel.
- Finding: the ensemble on raw items outperforms both benchmarks on AUC and especially at the top of the list (NDCG@1%).
- Why it matters: a regulator or auditor can only investigate a few firms; ranking quality at the top is what counts.

Literature review: data and labels

Fraud labels (AAERs)

SEC Accounting and Auditing Enforcement Releases compiled by UC Berkeley CFRM (May 1982 to Sep 2016), extended by the authors by hand to Dec 2018 (AAER #4012). A firm-year is "fraud" if it is covered by an AAER. Serial frauds share one case id (p_aaer).

Financial data (Compustat)

Annual fundamentals, fiscal years 1991-2014. 146,045 firm-years from 18,444 firms; 964 fraud firm-years in 412 AAER cases, a base rate of 0.66%.

28 raw items

act, ap, at, ceq, che, cogs, csho, dlc, dltis, dltt, dp, ib, invt, ivao, ivst, lct, lt, ni, ppegt, pstk, re, rect, sale, sstk, txp, txt, xint, prcc_f

14 financial ratios (Dechow et al.)

dch_wc, ch_rsst, dch_rec, dch_inv, soft_assets, ch_cs, ch_cm, ch_roa, issue, bm, dpi, reoa, EBIT, ch_fcf. The paper shows raw items beat these ratios, so the hand-made features of prior work were not helping the model.

Literature review: protocol and model

1	**Expanding window**	Train on 1991 to T-2, test on year T, for T = 2003 to 2014. Mimics how a regulator would use it.
2	**2-year gap**	Fraud in year T-1 is usually not public yet, so T-1 is dropped from training. Prevents leakage of undiscovered fraud.
3	**Serial-fraud masking**	A fraud case that continues into the test year is removed from training so the model cannot "remember" the firm.
4	**Tuning**	Train 1991-1999, validate on 2001 (gap kept). Grid over number of learners 100 to 3000, chosen by AUC.
5	**RUSBoost**	300 boosted trees, MinLeafSize 5, learning rate 0.1, under-sample non-fraud to 1:1 each round (MATLAB fitensemble).

Literature review: metrics and findings

Metric	What it measures	Why the paper uses it
AUC	Probability a random fraud ranks above a random non-fraud	Standard, threshold-free
NDCG@k	Discounted hits in the top k% of the ranking	Investigators only look at the top of the list
Sensitivity@k	Share of all frauds caught in the top k%	Recall at a budget
Precision@k	Share of the top k% that are fraud	Hit rate per investigation

Limitations the authors acknowledge: only SEC-caught fraud is labelled (undetected fraud = 0); labels keep arriving for years; the model ranks, it does not decide.

Reported findings (approx., Table 3)

RUSBoost on 28 raw items: AUC about 0.72 over 2003-2008, ahead of the logistic F-score (about 0.69) and SVM-FK (about 0.63).

The raw-item model also wins on NDCG@1%, i.e. at the very top of the list.

Results are robust to a 1-year gap and to alternative test windows.

Our reproduction of the baseline: AUC 0.705 (paper tree settings) to 0.724 (deeper trees) over 2003-2014, consistent with the paper.

2 · Research gaps

What the paper leaves open, found by re-implementing it line by line

Five gaps we set out to close

1. Scale bias

Raw dollar items (assets, shares, share price) make the model partly learn "big firms that issue equity". Risk is not scale-free.

2. No explanation

RUSBoost is a black box of 300 trees. An auditor cannot see WHY a firm is flagged.

3. No decision step

The paper ranks firms. There is no cutoff, no confusion matrix, no calibrated probability an investigator can act on.

4. Thin significance

Model differences are reported without confidence intervals; fraud cases span years, so observations are not independent.

5. Not deployable

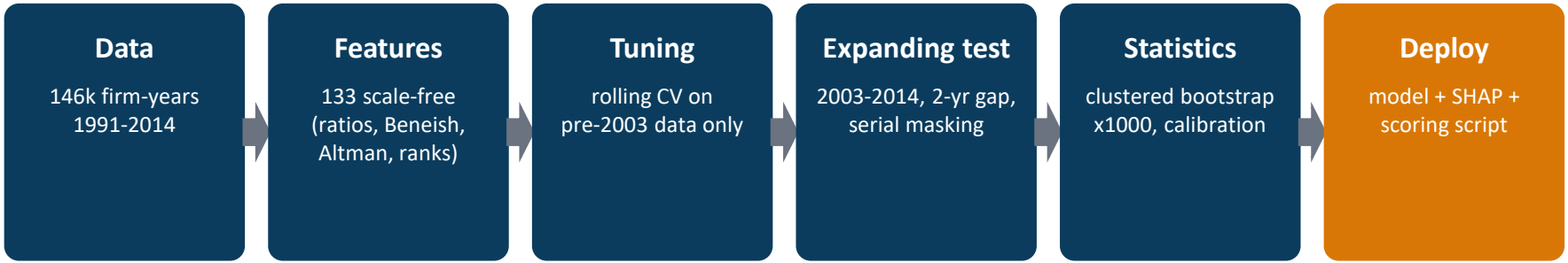
No scoring path for new firms; later training years have incomplete labels (fraud is found years later).

Each gap maps to one implemented improvement on the next slides.

3 · What we implemented

Same protocol as the paper, so every number is comparable

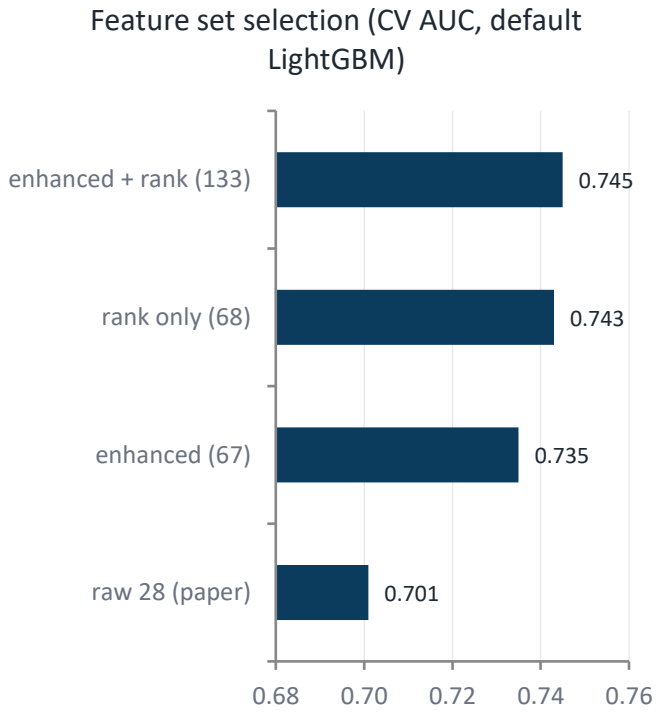
Pipeline overview



- Everything the paper did is kept: expanding window, 2-year gap, serial-fraud masking, tuning only on data before 2003.
- Added: winsorising refit on every training window (no fixed 1991-2001 caps leaking into 2014); test years with under 15 frauds excluded from averages (2013, 2014).
- Baseline re-implemented in Python from the authors' MATLAB settings (300 trees, leaf 5, LR 0.1, 1:1, 10 splits per tree) and reproduces their AUC range.
- Code: run_pipeline.py (18 min on a laptop CPU), src/{features, models, evaluate, explain}.py, score_new_data.py, smoke test.

Improvement 1: scale-free feature engineering

- Dropped all 28 raw dollar items from the proposed model (kept for the baseline only).
- Added Beneish M-score components (DSRI, GMI, AQI, SGI, DEPI, TATA, LVGI), Altman Z, common-size balance-sheet ratios, growth gaps, external-financing need.
- Added within-year percentile ranks of every ratio: a firm is compared with its own year's market, which removes era drift (1991 vs 2014).
- Feature set chosen by validation AUC on pre-2003 folds, never on test years.



Improvement 2: tuned, balanced-bagged ensemble

- Members: LightGBM, XGBoost and logistic regression, each trained as balanced bags (all frauds plus a random 5-10x sample of non-frauds, 10 bags).
- Hyper-parameters by random search (20 trials each) on rolling-origin CV using only pre-2003 data.
- Members are rank-averaged within each test year, exactly as the paper scores within a year.
- Ensemble chosen on validation AUC with a simplicity tie-break; the baseline model is never a member.

Candidate	CV AUC
RUSBoost raw 28 (baseline)	0.690
RUSBoost enhanced	0.736
LightGBM	0.746
XGBoost	0.749
Logistic regression	0.743
LGBM + XGB	0.751
LGBM + XGB + LR (chosen)	0.760

Improvements 3 to 5: decision step, statistics, deployment

3. Decision step

Precision, recall and lift at a 1%, 2% and 5% investigation budget, per year, plus a confusion matrix at 1%.

Isotonic regression turns the rank score into a calibrated fraud probability (fit on out-of-sample 2003-2010 scores).

Plain-language output: "frauds found per 100 investigations".

4. Honest statistics

Paired bootstrap, 1,000 resamples, clustered by AAER case for frauds and by firm for non-frauds (observations are not independent).

CIs for AUC, NDCG@5% and Sens@5%, not just AUC.

Years with fewer than 15 frauds (2013: 11, 2014: 4) are listed but excluded from averages.

5. Deployment and SHAP

Deployable model trained on 1991-2010 only, because 2011+ labels are still arriving (69 frauds in 2003 vs 4 in 2014).

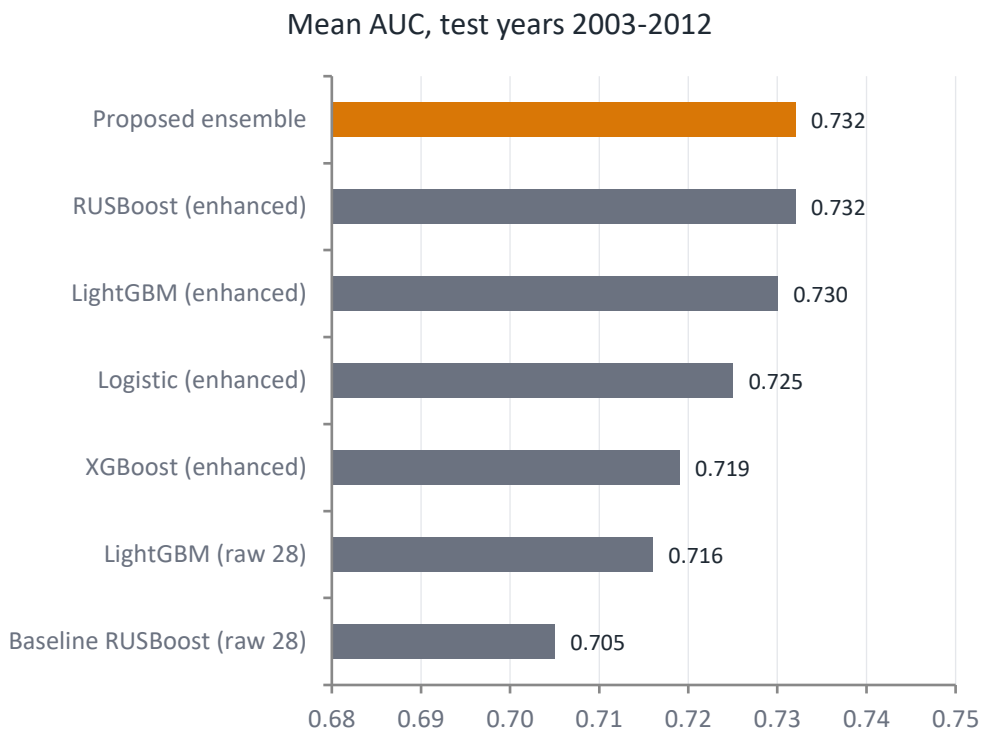
SHAP computed on the actual deployed members, all bags, on a real-base-rate sample of 2012-2014.

score_new_data.py ranks new firms against the full market of the same year and outputs probability and percentile.

4 · Results

Test years 2003-2012 (2013-2014 excluded for too few frauds); mean of yearly metrics; same protocol as the paper

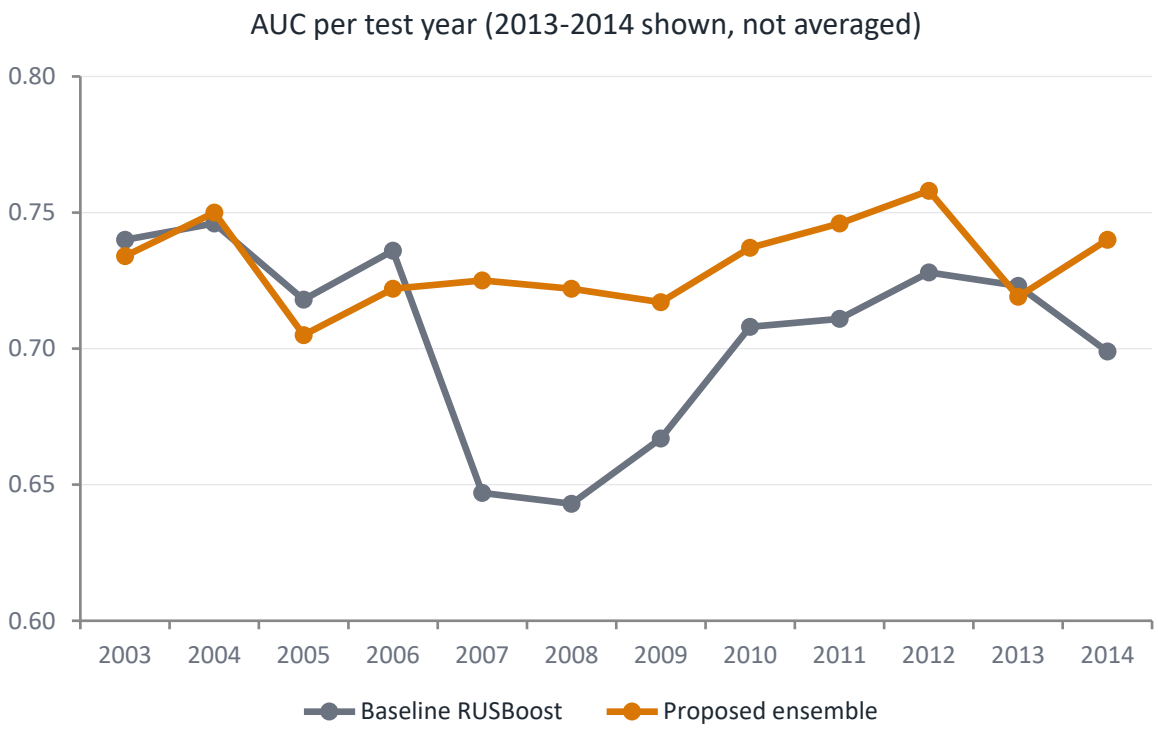
Headline: AUC 0.705 to 0.732, a significant gain



Metric	Gain	95% CI	P(gain<= 0)
AUC	+0.027	[+0.007, +0.045]	0.002
Sens@5%	+0.057	[+0.013, +0.095]	0.004
NDCG@5%	+0.015	[-0.012, +0.042]	0.135

- AUC and Sens@5% gains are statistically significant under a case-clustered bootstrap.
- NDCG@5% improves but not significantly.
- RUSBoost on the new features alone reaches 0.732: the features do most of the work, the ensemble adds top-of-list recall.

Year by year: the gain is consistent after 2006



- Baseline collapses in 2007-2009 (AUC 0.64-0.67, the financial-crisis years); the proposed model stays near 0.72.
- Proposed wins 7 of the 10 scored years; the losses (2003, 2005, 2006) are within 0.015.
- Fraud counts per test year fall from 69 to 4, which is why late years are noisy and why the deployable model stops training at 2010.

Decision step: what an investigator actually gets

1.9
frauds per 100 investigations (top 1%), proposed

1.8
baseline RUSBoost

0.6
random picking (base rate)

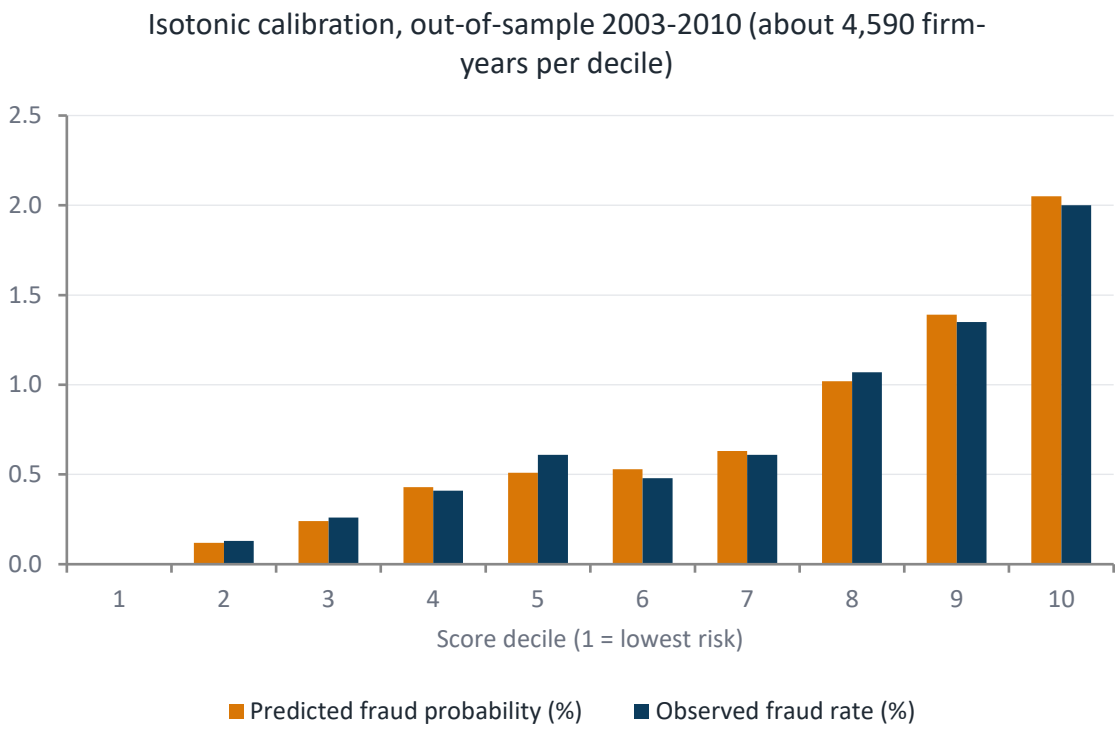
3.5x
lift over random

Budget	Precision	Recall	Lift
Top 1%	1.9%	3.6%	3.5x
Top 2%	2.0%	6.5%	3.2x
Top 5%	2.0%	15.5%	3.1x

2004, top 1% (60 firms)	Fraud	Not fraud
Flagged	TP 3	FP 57
Not flagged	FN 55	TN 5,819

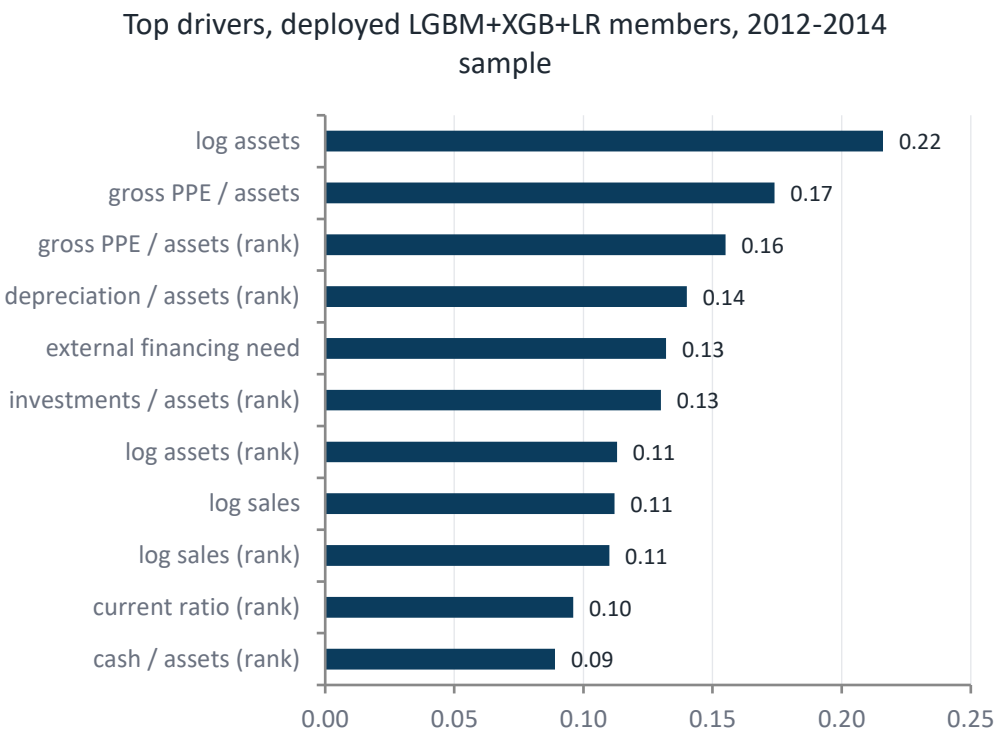
Honest framing: about 98 of every 100 flagged firms are not (labelled) fraud. This is a prioritisation tool that triples the hit rate of an investigator, not a fraud detector. Several years have zero hits at 1%.

Calibrated probabilities: the score means what it says



- Predicted and observed rates match in every decile; top decile 2.1% predicted vs 2.0% observed.
- Bagging at 5:1 / 10:1 makes raw model outputs 10-100x too high; calibration fixes that before anyone reads them as probabilities.
- Limit: it is a step function fit on 8 years and saturates near 2-3%.

Explainability: SHAP on the deployed model



- Drivers an auditor recognises: heavy fixed assets, low depreciation, high external financing need, thin liquidity.
- Each flagged firm gets a waterfall plot: which features pushed its score up or down.
- Honest caveat: firm size (log assets, log sales) still dominates even without raw dollar items. Dropping size would make the model size-neutral but weaker.
- SHAP explains what the model uses, not why a firm committed fraud; the top-flagged table shows real hits and false alarms side by side.

Confident lines to present

"We reproduced the JAR 2020 baseline in Python with the authors' exact settings, and it lands in their reported AUC range."

"Replacing raw dollar items with 133 scale-free accounting features lifts AUC from 0.705 to 0.732 and top-5% recall from 9.8% to 15.5%, both significant under a case-clustered bootstrap."

"Unlike the paper, we give an investigator a decision: at a 1% budget the model finds 1.9 frauds per 100 reviews, 3.5x random, with a calibrated probability behind every score."

"Every flagged firm comes with a SHAP explanation of the deployed model itself, not a proxy model."

"The deployable model stops at 2010 on purpose: fraud labels for later years are still being discovered."

5 · What a major project could add

Suggestions only, none implemented. Ordered by expected value per week of work.

Realistic extensions for a major project

Text features from 10-K filings

MD&A tone, readability, length changes (Loughran-McDonald dictionaries or a small fine-tuned language model). Published work shows text adds signal beyond the financials.

Size-neutral model

Drop log assets / log sales, or model fraud risk within size buckets, and test whether precision at 1% holds. Decides whether "size" is signal or bias.

Cost-sensitive cutoff

Pick the investigation budget by expected value: cost of a review vs. loss avoided per catch. Turns the ranking into a defendable policy.

Label-delay modelling

Weight recent training years by the probability their frauds are already discovered (survival-style), instead of a hard 2010 cutoff.

Stronger statistics

DeLong test per year, firm-level cross-validated calibration, and a stricter nested selection so the 3 validation years cannot overfit.

Other markets

Re-run on an Indian or EU sample (different standards, different fraud patterns). The README already anticipates a 300-company Indian dataset.

Thank you

Code, results and this deck: Final-Year-Project repository

run_pipeline.py reproduces every number in 18 minutes on a laptop; score_new_data.py scores new firms.